What Does 44 % Mean? Constant Baselines for Tooth-Level Caries Evaluation on Panoramic Radiographs

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***Abstract*—Automated screening of panoramic radiographs is almost always reported with tooth-level metrics: which teeth are carious. Such numbers are easy to compute, easy to compare, and—we argue—remarkably hard to interpret, because the literature rarely establishes what an uninformed system would score on the same images. We built a six-stage panoramic pipeline, validated it on four external hospitals, calibrated its confidences and gave it a principled way to decline, and then set out to answer a narrower question: what does its caries number actually mean? To find out, we scored it against progressively less informed opponents under one identical scorer—a 33 B-parameter vision–language model, an 8 B one, and finally a baseline that is not shown the radiograph at all. The comparison did not resolve the way we expected, and it did not resolve the same way inside and outside our development data. We report what separates a diagnostic system from an anatomical prior on this task, what does not, and which of our own results survive the distinction.**

***Index Terms*—panoramic radiography, dental caries, evaluation methodology, baselines, external validation, model calibration, selective prediction**

I. Introduction

Consider a screening report. Against a panoramic radiograph it lists four tooth codes—16, 26, 36, 46—and marks them carious. A clinician reading that report has a reasonable first impression: the system has looked at the image and found something. The codes are specific. They are plausible. They are, often enough, correct.

Now consider a second program. It emits the same four codes for every radiograph it is ever shown. It contains no network, no features, no pixels— only a list. On a large enough set of panoramic images, it too will be right often enough to produce a respectable-looking number.

The distance between those two programs is the subject of this paper.

Panoramic radiographs (OPGs) are the most common extraoral radiograph in dentistry, and automated caries and periodontal screening on them is an active field [1]–[3]. Reported tooth-level accuracies cluster in a range that reads as clinically useful. What is much rarer is a statement of the floor: what the same metric would return for a system that does not perceive anything at all.

Two properties of this task make that floor unusually high, and both are anatomical rather than algorithmic. First, caries is not uniformly distributed across the dentition—it concentrates heavily in molars. Second, tooth-level metrics ask a *set-membership* question: which teeth, from a fixed vocabulary of 32 codes. A question of that shape can be answered from anatomy alone, without reference to any particular patient. Together these mean that a fixed answer inherits a substantial score by construction.

The natural response is to check. We surveyed 49 task-relevant papers from our own review of the area; **two** report validation on data from an institution other than the one that produced their training set, and **none** report a constant, image-blind, or otherwise uninformed baseline. The floor is, in this literature, essentially unmeasured.

This paper is our attempt to measure it on our own work rather than anyone else's. We first built a system worth testing—a six-stage pipeline performing tooth detection, FDI numbering, caries classification and periodontal bone-loss classification, evaluated with cluster-bootstrapped confidence intervals, calibrated probabilities, a pre-registered abstention threshold, and external validation across four hospitals it never trained on. We then constructed a ladder of opponents, each given strictly less information than the last, and scored every rung with one scorer on one set of images.

We report the outcome in full, including the parts that do not favour our system, and including a disagreement between our development data and our external data that we did not anticipate and cannot yet explain. Our contributions are:

1) A **constant-baseline protocol** for tooth-level OPG evaluation, together with what it reveals when applied across two unrelated metric families and five hospitals.
2) A **vision–language-model baseline with the controls that make it interpretable**—including what these models answer when shown an image containing no anatomy whatsoever.

3) **External validation on four hospitals**, reported per site, with the spread that pooling conceals.
4) **Calibration and selective prediction** results, each with a control isolating the mechanism responsible.
5) An honest characterisation of the pipeline itself, including one branch that does not survive domain shift and a headline number that is bounded by an earlier stage.

We are explicit about what we do not claim. We do not assert that published caries results are incorrect. We argue something narrower and, we think, more useful: that a tooth-level number reported without a floor cannot be distinguished from an anatomical prior—a statement about *evidence*, not about anyone's system.

II. Related Work

A. Detection and classification on panoramic radiographs

Deep learning on OPGs has converged on a familiar decomposition: detect teeth, then classify each one. Detector choice is well studied, and recent head-to-head evaluation on dental panoramics supports YOLOv11 over its successor generation for multi-class dental detection [4], [5]. The same body of work consistently reports an asymmetry that matters here: performance is strong on well-defined objects (crowns, implants, restorations) and markedly weaker on subtle radiolucent pathology such as caries and bone loss [6], [7]. Caries segmentation and classification specifically have been approached with dual-network frameworks [1], attention architectures [3], and cascaded designs [2].

B. Tooth numbering

FDI numbering is typically reported at very high accuracy—frequently above 0.97—but often on *pre-cropped* teeth, where the task is a 32-way classification of an image that is already known to contain exactly one tooth, correctly framed [8], [9]. This is a materially easier problem than numbering under detector output, where teeth may be missed, duplicated or clipped. We report both framings and find the gap between them large.

C. Vision–language models in dentistry

Recent work applies general-purpose and domain-adapted VLMs to dental imaging [10]–[13]. These systems are attractive because they need no task-specific training, and are difficult to evaluate for the same reason: a fluent, confident, correctly formatted answer is not evidence of perception. We adopt a control from the broader shortcut-learning literature [14]: present the model with an input containing none of the signal it purports to read, and observe whether its behaviour changes.

D. External validation and the missing floor

External validation remains uncommon. In our review of 49 task-relevant papers, two evaluate on an outside institution. Where it is done, degradation is often severe—one reported system falls from 95.06 mAP on its own data to 61.83 on a public benchmark. Independent work on periodontal bone loss reports weighted F1 of 0.29 with severe-class F1 of zero against clinician reference [15], [16], a reminder that in-domain figures can be a poor guide to behaviour elsewhere. To our knowledge, no work in this area reports an image-blind baseline, which is the gap this paper addresses.

III. System

The finding in this paper is only interesting if the system it is measured against is a serious one. We therefore describe the pipeline before the evaluation that tests it.

A. Architecture

The pipeline maps an OPG to a structured per-tooth record in six stages. **Stage 1** detects teeth with a YOLOv11l segmentation model [17]. **Stage 2** assigns each detection a two-digit FDI code using a ResNet18 [18] head that consumes the tooth crop together with four positional features, resolved to a one-to-one assignment across the mouth by the Hungarian algorithm [19]. **Stages 3 and 4** classify caries and periodontal bone loss per tooth using independent ResNet18 branches, each arranged as a coarse-to-fine cascade. **Stages 5 and 6** fuse the per-stage outputs on the FDI key and emit the record.

The whole system totals approximately 25 M parameters and runs at 2.39 s per image on CPU with no GPU present. This figure becomes relevant in Section V.

B. Data

Tooth detection and bone loss use the Tufts Dental Database [20] (1,000 OPGs) under a frozen 700/150/150 split. Caries uses DENTEX [21]. FDI numbering trains on both, with a third public corpus used for an ablation.

All splits are **image-level**. Two crops from one radiograph are not independent observations, and a tooth-level split would place the same mouth in both training and test data.

Two labelling conventions were established by measurement rather than assumed, because a silent error in either would have trained a plausible but wrong model. Tufts numbering is left–right **mirrored** relative to Universal notation (Tufts "1" corresponds to FDI 28, not 18), confirmed three independent ways. The third corpus was checked the same way: using the two-corpus numbering head as an independent arbiter over 915 teeth, agreement was **80.3 %** reading its labels as written against **1.7 %** reading them mirrored.

C. External data

Four hospitals from an independent Chinese cohort (denoted test1–test4) serve as locked external test data: never trained on, never tuned against, and evaluated a bounded and recorded number of times.

D. Crop geometry and pre-registration

Classifier inputs are expanded crops resized to 224×224. Expansion is fixed per branch and identical across corpora—symmetric $1.5\times$ for numbering, $1.5\times$ width by $2.0\times$ height for bone loss, since teeth are roughly three times taller than wide and horizontal over-expansion bleeds into neighbouring teeth.

Operating points are fixed *a priori* and reported where they fall rather than tuned against evaluation data. One consequence is reported here rather than omitted: the detector's mask-quality bar was pre-registered at 0.65 and **missed**, at 0.638. The bar was not moved after the fact; the model was adopted for its bounding boxes alone by an explicit, dated decision, and the mask-dependent component it was intended to enable remains unwired.

IV. Evaluation Protocol

Having built the system, we wanted to know what its caries number meant. The protocol below is the paper's methodological core.

A. The ladder

Four systems answer one question—*which teeth in this radiograph show caries?*—on the same 164 leakage-free DENTEX images, scored by one scorer per metric family. Each rung is given strictly less information than the one above.

1) **Pipeline**: the six stages of Section III.
2) **VLM (33 B)**: Qwen2.5-VL-33B-Instruct [22], zero-shot.
3) **VLM (8 B)**: the same family at 8 B.
4) **Constant**: a fixed set of FDI codes emitted for every image. **The radiograph is never read.**

B. Fairness

A rigged baseline is worse than no baseline. The VLMs are given the FDI numbering scheme in the prompt, told explicitly that teeth may be absent, and parsed permissively, so that neither notation nor formatting is scored as clinical error. Our pipeline receives no corresponding assistance. Every affordance favours the opponent.

The constant is fitted greedily on 78 leakage-excluded images that are disjoint from the scored set—overlap asserted to be zero at runtime—and then frozen. We also report a variant fitted directly on the test set; it is labelled an oracle throughout and is not claimable.

C. Metrics

We use two metric families, chosen to fail differently.

FDI-set F1 treats each image as a set-prediction problem: micro-averaged precision, recall and F1 over predicted versus annotated FDI codes.

Lesion-localisation agreement asks, of every expert-annotated lesion, whether a tooth was flagged that spatially overlaps it. It requires spatial correspondence, which set membership does not. Because it has no negative class by construction—a deliberate property, since our external cohort lacks healthy controls—it is a recall-style measure, and flagging more teeth can only raise it. We therefore always report it alongside the predicted-positive match rate and the number of teeth flagged, and we include an upper-bound arm that flags *every* detected tooth.

TABLE I
Tooth-level caries prediction, 164 leakage-free DENTEX images, 477 affected teeth. One scorer for all arms.

System	P	R	F1
Constant (image never read)	0.218	0.600	**0.320**
Our pipeline	0.235	0.390	0.294
Qwen2.5-VL-33B, zero-shot	0.250	0.243	0.246
Qwen2.5-VL-8B, zero-shot	0.248	0.107	0.149
Oracle constant (test-fitted)	—	—	*0.330*

TABLE II
Lesion-localisation agreement, same images, 479 annotated lesions. Teeth flagged is reported because agreement alone rewards flagging more.

System	Agreement	Teeth flagged	Match rate
Constant	**0.488**	859	0.272
Our pipeline	0.443	755	0.281
All teeth (ceiling)	*0.722*	*3,344*	*0.103*

D. Controls

For the VLMs, we present a **blank grey image** and a **random-noise image** under the identical prompt. Neither contains a tooth. For selective prediction, we **shuffle the confidences**, breaking the link between confidence and correctness while preserving both marginals; a gain that survives shuffling is an artefact of discarding items rather than of ranking them.

V. Results

A. The ladder, on development data

Fig. 1 summarises the comparison; Table I gives the set-prediction result exactly.

The ordering is inverted with respect to the information each system receives. The best-performing entry is the one that never sees a radiograph.

A single metric behaving oddly is a metric artefact. We therefore repeated the comparison under lesion localisation, which demands spatial correspondence (Table II).

The constant leads again, by **4.59 percentage points**, 95 % CI $[+0.60, +8.89]$ under a paired bootstrap over images. It does not achieve this by indiscriminate flagging: it flags 14 % more teeth than the pipeline at an essentially identical match rate. Two metric families designed to measure different things therefore agree.

B. The ladder, on external data

We then ran the same comparison on four hospitals the system had never seen (Table III).

The development-set result does not replicate. Pooled, the difference is -0.89 percentage points with 95 % CI $[-4.88, +3.01]$: the two systems are statistically indistinguishable, with the pipeline nominally ahead by an amount that is indistinguishable from noise.

The per-site spread is the more useful observation. The difference swings from $+15.8$ points at test1 to -10.6 at test2—a 26-point range across four hospitals of a single cohort—

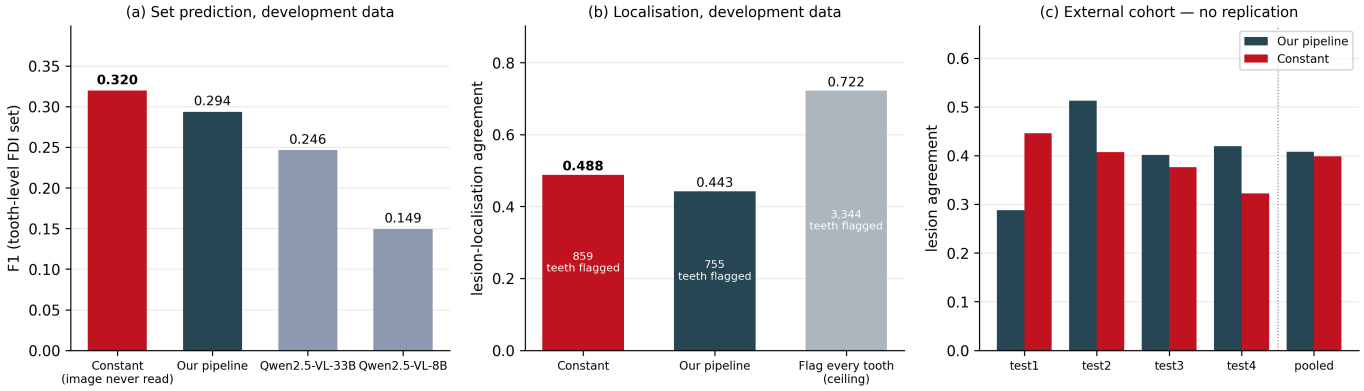


Fig. 1. The constant-baseline argument. **(a)** Set-prediction F1 on 164 leakage-free development images: the ordering is inverted with respect to the information each system receives, and the best-scoring entry never reads a radiograph. **(b)** A second, unrelated metric requiring spatial correspondence returns the same ordering; teeth flagged is annotated inside each bar because agreement is recall-shaped, and the flag-everything arm shows what saturating the metric costs. **(c)** The external cohort, where the result does *not* replicate: the difference swings 26 points between sites and vanishes when pooled. Every value is read from the saved evaluation artefacts at render time rather than transcribed.

TABLE III
EXTERNAL VALIDATION, FOUR HOSPITALS, 337 IMAGES, 1,013 LESIONS. CARIES LESION AGREEMENT BY CENTROID CONTAINMENT.

Hospital	Lesions	Pipeline	Constant	Δ
test1	316	0.288	**0.446**	+15.8
test2	349	**0.513**	0.407	−10.6
test3	162	0.401	0.377	−2.5
test4	186	0.419	0.323	−9.7
Pooled	1,013	0.408	0.399	−0.89

and the pooled figure conceals this entirely. Notably, test1 is simultaneously the pipeline's worst site and the constant's best. We do not have an explanation for that, and we report it unexplained rather than omit it.

Across all five hospitals, then, the defensible claim is *non-separability*: a purpose-built pipeline and an image-blind prior cannot be reliably told apart by these metrics. That is weaker than the development-set result alone would support, and it is the claim the full evidence sustains.

C. What the constant cannot do

A constant answers a set-membership question. It cannot perform tasks whose output is not a fixed list, and the pipeline's results on those tasks are correspondingly meaningful.

Tooth numbering. End-to-end FDI-exact accuracy is 58.3 % (546/936 teeth), 95 % CI [53.9, 62.8]. This figure is **detection-bound** and we report it as two numbers: 29.4 % of ground-truth teeth are never detected, and of the 661 teeth Stage 1 does find, Stage 2 numbers **82.6 %** correctly. Residual errors are largely benign: of 115 incorrect codes, 89 (77.4 %) are off-by-one within the correct quadrant, and only 16 of 661 detected teeth (2.4 %) receive a quadrant-level error. A constant cannot attempt this task at all.

Calibration. The caries gate's confidences are not probabilities: expected calibration error is 0.136 and the model is systematically overconfident. Temperature scaling [23] made calibration *worse* (0.152), indicating the miscalibration is structural rather than a single scale factor; isotonic regression [24] reduced ECE to **0.043**, a 68 % improvement, and is applied to confidences only, leaving every verdict byte-identical.

Selective prediction. Declining the least-confident quarter of cases raises numbering accuracy from 0.826 to **0.885** (+5.91 pp, CI [+3.88, +7.68]) and caries-gate accuracy from 0.797 to **0.940** (+14.25 pp, CI [+11.32, +16.63]). Under the shuffle control both gains collapse to +0.03 and +0.00 percentage points respectively, establishing that the improvement comes from the confidence ranking and not from discarding items. The pre-registered threshold of 0.30 lands at 90.8 % coverage and 0.847 accuracy, within 0.2 points of the empirical 90 %-coverage operating point—a value chosen before these data were seen.

One caveat must accompany any coverage figure here: coverage is measured over *detected* teeth. Retaining 75 % of detections corresponds to 75 % of 70.6 %, or 53.0 % of the teeth actually present.

D. Vision–language model behaviour

Few-shot prompting, with exemplars drawn strictly from leakage-excluded images, degrades the 8 B model monotonically: F1 falls from 0.149 zero-shot to 0.066 at two shots and 0.034 at four. The mechanism is abstention rather than error— empty answers rise from 56 of 164 images to 123 of 164—so exact-set match simultaneously *rises*, because 16 images contain no caries and an empty answer scores an exact match on each. Reported alone, that column would be true and thoroughly misleading.

The control is more direct. Presented with a **blank grey image** and with **random noise**, both models return confident, correctly formatted, dental-specific answers; for one model the response to pure noise is identical to its response to a

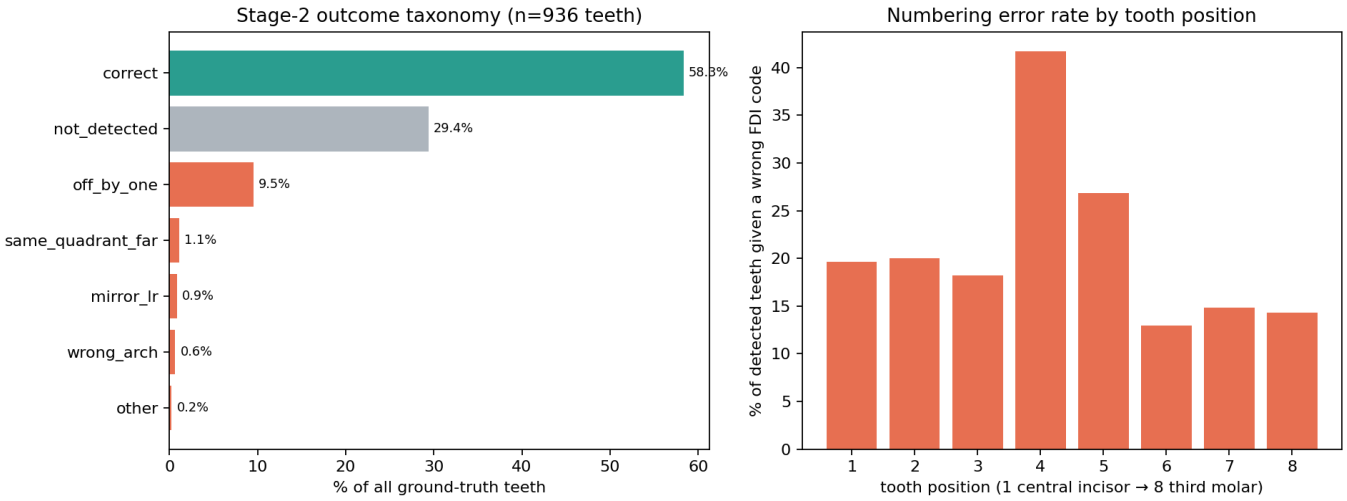


Fig. 2. Where the numbering stage loses teeth. The largest single category is teeth the detector never proposes, which no numbering model can recover; among the codes that are wrong, most are off-by-one within the correct quadrant.

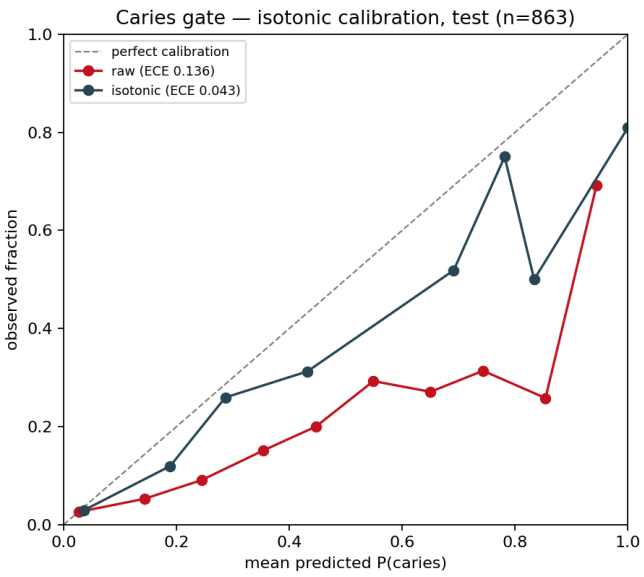


Fig. 3. Reliability of the caries gate before and after isotonic regression. The uncorrected model is systematically overconfident. The correction is applied to confidences only, leaving every verdict unchanged.

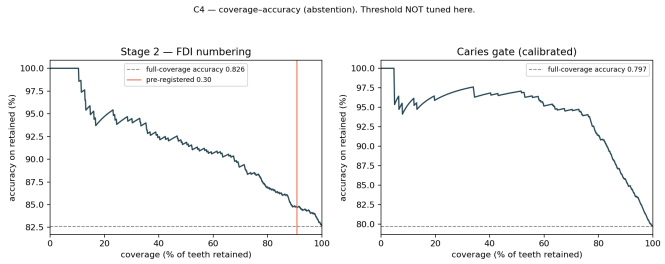


Fig. 4. Coverage–accuracy for tooth numbering (left) and the caries gate (right). Accuracy on retained predictions rises as the least-confident cases are declined. The vertical line marks the pre-registered threshold of 0.30, located on the curve rather than tuned against it.

real radiograph. Whatever these outputs measure, it is not perception of the image in front of them.

The efficiency comparison survives all of the above and is worth stating plainly: 33 B parameters and 70 GB of accelerator memory are outperformed by approximately 25 M parameters running on a CPU.

E. Where the pipeline fails

The bone-loss branch reaches 0.923 macro-F1 in-domain on 1,558 validation crops at an untouched 4 % prevalence (sensitivity 83.05 %, specificity 99.53 %) and **does not survive domain shift**, firing on 0.7 % of DENTEX teeth and reaching only 1.4–5.6 % affected-tooth sensitivity across the four external hospitals. A detector change that moved numbering by +12.0 points and caries agreement by +7.7 moved bone loss by nothing, which falsifies the hypothesis that a shared upstream cause explains all three. We report this as a measured negative result rather than a component.

VI. Discussion

A. Why the floor is high

Caries in these cohorts is anatomically concentrated in molars. A tooth-level decision is a set-membership question, and set membership inherits that base rate directly. The consequence is that headroom above a fixed anatomical answer is small—and, importantly, small *independently of model quality*. This is a property of the task–metric pairing, not of any particular system.

That two unrelated metrics agree on development data matters. Set-membership F1 and lesion localisation were designed to measure different things; the second was adopted specifically because it requires spatial correspondence. A single-

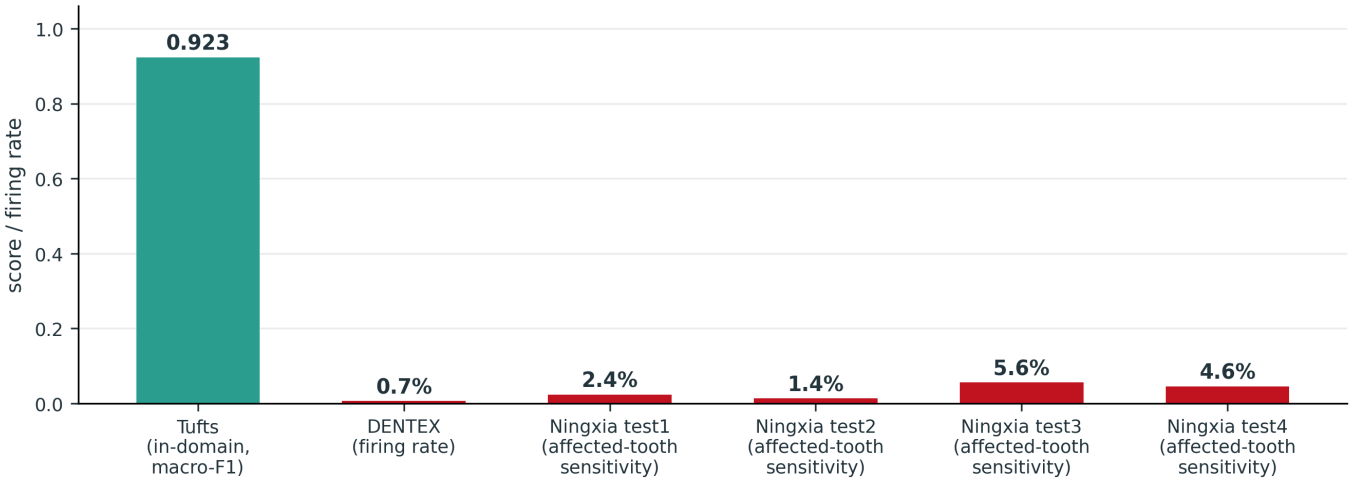


Fig. 5. The bone-loss branch under domain shift. In-domain macro-F1 of 0.923 on Tufts does not transfer: the branch fires on 0.7 % of DENTEX teeth and reaches 1.4–5.6 % affected-tooth sensitivity across the four external hospitals. Metrics necessarily differ between cohorts—the external sets are scored by affected-tooth sensitivity—but the collapse does not depend on that choice. Reported as a measured negative result rather than a component.

metric result would be a metric artefact. Two are a property of the data.

B. Why the external disagreement matters more

We expected the external cohort to confirm the development finding. It did not: pooled, the systems are indistinguishable, and per site the difference swings through 26 points. Two readings are available and we cannot presently separate them. Either the development result is partly a property of DENTEX's particular caries distribution, or the external metric—which necessarily uses centroid containment rather than overlap, because external lesions are small focal blobs inside much larger tooth boxes—has different sensitivity. Both are consistent with the data we have.

What the disagreement does establish is that a *single* external number is not informative at this variance. Our own pooled figure of 0.408 is, on its own, as uninformative as any other pooled figure in this literature. We report per site for that reason, and recommend the same of others.

C. What should be reported

We propose that a constant-baseline row accompany any tooth-level caries metric on this class of data. It costs one line of code and no compute. Without it, a mid-range tooth-level number cannot be interpreted—as our four systems demonstrate, spanning 0.149 to 0.320 with the *least* informed entry at the top.

This does not invalidate prior work. It means one specific claim—*this system perceives caries*—is not established by a tooth-level number alone.

D. What still separates a system from a prior

A constant cannot number teeth, cannot produce a calibrated confidence, cannot abstain informatively, and has no behaviour under domain shift to measure. We propose these as the axes on which OPG systems should be compared:

1) **Identification**: can it say *which* tooth? (82.6 % of detected teeth here.)
2) **Calibration**: are its confidences probabilities? (ECE 0.136 $\rightarrow$ 0.043.)
3) **Selective prediction**: does declining improve accuracy, and does the gain survive a shuffle control? (+14.25 pp; yes.)
4) **Domain shift**: measured on held-out institutions and reported per site.

Our pipeline performs meaningfully on all four. Its caries F1 conveys much less.

E. On reporting negative results

Two findings here are unfavourable to our own system: the constant comparison, and a bone-loss branch that does not generalise. We report both, because a screening tool that fails silently is more dangerous than one that fails visibly—and because the alternative, selecting whichever metric flatters the system, is precisely the practice this paper argues against.

VII. Limitations

1) **Two caries cohorts, disagreeing in direction.** The constant leads on development data and ties externally. We report both rather than the more striking one; the common finding is non-separability.
2) **No clinician comparison.** We do not know how any system here compares against a dentist on these images. This is the single largest gap and is the subject of ongoing work.
3) **Tooth-level is our finest localisation.** The localisation metric is a tooth-level correspondence check, not a pixel-level segmentation assessment.
4) **One VLM family** at two scales; a domain-adapted medical VLM was not accessible.

5) **Detection bounds Stage 2.** At 0.810 box mAP@0.5, 29.4 % of teeth are never presented to the numbering head.
6) **The external cohort lacks healthy controls**, so specificity and AUC are not computable there; we report lesion agreement only.

VIII. Conclusion

We set out to determine what a tooth-level caries number means, using our own system as the subject. On development data, a baseline that never reads the radiograph outscored a purpose-built pipeline and two vision–language models under two unrelated metrics. Across four external hospitals the same comparison produced no separation at all, with a 26-point swing between sites.

The conclusion we draw is not that automated screening does not work. It is that this measurement cannot tell us whether it does. Identification, calibration, selective prediction and behaviour under domain shift can—and on those axes, reported here with their controls, our pipeline has something to show. We recommend that tooth-level caries results be accompanied by an image-blind baseline and reported per institution, so that the next number of this kind is interpretable when it appears.

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